Customer Churn Prediction Platform

Digital Egypt Pioneers Initiative (DEPI) — Graduation Project Documentation

TRACK & PROGRAM	REPOSITORY NAME	DOCUMENT VERSION
AI & Data Science (DEPI)	customer-churn-retention	Part 1: Initial Deliverable

PART 1: PROJECT PLANNING & MANAGEMENT

Customer churn presents a critical financial challenge for subscription-based enterprises. Acquiring new subscribers costs significantly more than retaining existing accounts. This project delivers an end-to-end Machine Learning and MLOps framework designed to quantify customer churn risk, analyze behavioral drivers, and provide actionable retention intelligence.

1.1 Project Objectives

- **Technical Objectives:** Implement, evaluate, and compare multiple classification algorithms (Logistic Regression, Decision Trees, KNN, SVM, Gradient Boosting) using rigorous validation protocols without data leakage.
- **MLOps & Tracking:** Integrate MLflow to systematically log hyperparameters, metrics, and model artifacts for 100% experiment reproducibility.
- **Business Impact:** Identify primary churn indicators to enable retention teams to prioritize high-risk, high-value customer accounts proactively.

1.2 Project Scope

- **In-Scope:** Data cleaning, Exploratory Data Analysis (EDA), pipeline feature engineering, model training/tuning, MLflow tracking, evaluation metrics, and unit testing strategy.
- **Out-of-Scope:** Live cloud deployment (Azure ML) and custom REST APIs (optional extensions depending on timeline reserve).

1.3 PROJECT TIMELINE & MILESTONES

Phase	Milestone / Task Description	Start Date	End Date	Status
Phase 0	Team Formation, GitHub Repository, DEPI Admin Setup	02-10-2026	08-10-2026	COMPLETED
Phase 1	Project Planning, Scope Definition & Documentation	02-10-2026	16-10-2026	COMPLETED
Phase 2	Literature Review & Requirements Gathering	09-10-2026	16-10-2026	COMPLETED
Phase 3	Data Understanding, Cleaning & Exploratory Analysis (EDA)	16-10-2026	23-10-2026	IN PROGRESS
Phase 4	System Analysis & Pipeline Architecture Design	09-10-2026	06-11-2026	SCHEDULED
Phase 5	Data Preparation, Feature Engineering & Pipelines	23-10-2026	06-11-2026	SCHEDULED

1.4 TEAM ROLES & RACI MATRIX

Role	Primary Responsibilities	RACI Designation
Project Manager / Lead	Timeline tracking, milestone management, DEPI LMS submissions, risk mitigation.	Accountable (A)
Data Engineer / Analyst	Data acquisition, schema validation, missing value handling, EDA reports.	Responsible (R)
Machine Learning Engineer	Model execution (Logistic Regression, Trees, Boosting), hyperparameter tuning.	Responsible (R)
MLOps & QA Lead	MLflow infrastructure, environment calibration, testing, repo management.	Responsible (R)
Technical Instructor	Guidance, compliance review, milestone approvals.	Consulted (C)

1.5 RISK MANAGEMENT & MITIGATION STRATEGY

Identified Risk	Impact / Prob.	Mitigation Strategy
Class Imbalance in Target	High / High	Evaluate via Precision, Recall, and F1-score rather than Accuracy alone. Resample on training split only.
Data Leakage in Pipelines	High / Med	Enforce strict train/test isolation using `scikit-learn` Pipelines prior to transformations.
Scope Creep (UI / Cloud)	Med / High	Freeze core requirements first; treat UI/Cloud endpoints as post-baseline optional extensions.
Version Inconsistencies	Med / Low	Lock dependencies in `requirements.txt` and enforce uniform Python environments.

PART 2: LITERATURE REVIEW

Customer Churn Prediction is a foundational problem in predictive analytics for subscription models (Telecom, SaaS, Banking). Historical approaches relied on static business heuristics (e.g., flagging accounts with zero activity for 30 days). Modern machine learning methodologies construct predictive classification workflows.

2.1 Existing Solutions Comparison

- **Statistical Baselines (Logistic Regression):** Offer high interpretability and low computational cost, but fail to capture non-linear feature interactions without manual feature construction.
- **Tree Ensembles (Random Forest, Gradient Boosting):** Consistently deliver superior performance on tabular business datasets by modeling complex conditional interactions automatically.

2.2 Solution Differentiation

Our platform advances standard Jupyter Notebook workflows by structuring code into modular Python packages, tracking all parameters and outputs via MLflow, and strictly separating train/test pipelines to prevent data leakage.

PART 3: REQUIREMENTS GATHERING

3.1 Stakeholder Identification

- **Primary Users:** Retention Operations Team, Customer Success Managers, Marketing Leads.
- **Technical Stakeholders:** Data Science Team, MLOps Engineers, DEPI Academic Evaluators.

3.2 Functional Requirements (FR)

- **FR-01 (Data Ingestion & Cleaning):** The platform shall ingest tabular datasets, validate schemas, remove duplicate entries, and impute missing values cleanly.
- **FR-02 (Exploratory Data Analysis):** The system shall generate visual and tabular distribution reports highlighting target churn correlation across customer segments.
- **FR-03 (Feature Engineering Pipeline):** The system shall automate categorical encoding (One-Hot / Target) and numerical scaling using isolated scikit-learn Pipelines.
- **FR-04 (Model Training & Selection):** The platform shall train baseline models (Logistic Regression, Decision Trees, KNN, SVM, Gradient Boosting) and report comparative performance.
- **FR-05 (MLflow Experiment Logging):** The system shall automatically log model artifacts, execution parameters, and performance metrics (Precision, Recall, F1, ROC-AUC) to an MLflow backend.
- **FR-06 (Model Persistence & Reloading):** The platform shall export serialized pipeline and model binaries for offline inference on new, unseen customer batches.

3.3 Non-Functional Requirements (NFR)

- **NFR-01 (Reproducibility):** The entire workflow must be executable end-to-end using a single entry command and a defined seed value.
- **NFR-02 (Maintainability):** Code must adhere to modular structures (separate modules for data loading, feature engineering, modeling, and evaluation).
- **NFR-03 (Security & Compliance):** Datasets must be fully anonymized with no raw personal identifiable information (PII) stored in the GitHub repository.

3.4 USER STORIES

> **User Story 1 — Retention Manager:** "As a Retention Specialist, I want to view a ranked list of high-risk customers alongside their primary churn indicators so that I can dispatch targeted retention offers effectively."

> **User Story 2 — Data Scientist / MLOps Lead:** "As a Data Scientist, I want every model training run and hyperparameter adjustment automatically logged in MLflow so that I can objectively justify final model selection to project reviewers."

> **User Story 3 — Technical Evaluator:** "As a DEPI Instructor, I want a well-structured GitHub repository with unit tests and reproducible execution guidelines so that I can assess individual team contributions accurately."